The optimal assignment (minimum) cost

= ₹ 38

Example 10.8

Consider the problem of assigning five jobs to five persons. The assignment costs are given as follows. Determine the optimum assignment schedule.

Person	Job 1	2	3	4	5
A	8	4	2	6	1
B	0	9	5	5	4
C	3	8	9	2	6
D	4	3	1	0	3
E	9	5	8	9	5

Solution:

Here the number of rows and columns are equal.

∴ The given assignment problem is balanced.

Now let us find the solution.

Step 1: Select a smallest element in each row and subtract this from all the elements in its row.

The cost matrix of the given assignment problem is

Person	Job 1	2	3	4	5
A	7	3	1	5	0
B	0	9	5	5	4
C	1	6	7	0	4
D	4	3	1	0	3
E	4	0	3	4	0

Column 3 contains no zero. Go to Step 2.

Step 2: Select the smallest element in each column and subtract this from all the elements in its column.

Person	Job 1	2	3	4	5
A	7	3	0	5	0
B	0	9	4	5	4
C	1	6	6	0	4
D	4	3	0	0	3
E	4	0	2	4	0

Since each row and column contains atleast one zero, assignments can be made.

Step 3 (Assignment):

Examine the rows with exactly one zero. Row B contains exactly one zero. Mark that zero by □ (i.e) PersonB is assigned to Job 1. Mark other zeros in its column by × .

Person	Job 1	2	3	4	5
A	7	3	0	5	0
B	[0]	9	4	5	4
C	1	6	6	0	4
D	4	3	0	0	3
E	4	0	2	4	0

Now, Row C contains exactly one zero. Mark that zero by □. Mark other zeros in its column by ×.

Person	Job 1	2	3	4	5
A	7	3	0	5	0
B	[0]	9	4	5	4
C	1	6	6	[0]	4
D	4	3	0	~~0~~	3
E	4	0	2	4	0

XII Std - Business Maths & Stat EM Chapter 10.indd 254 2/29/2020 11:45:21 AM